The Stanat-Weiss Pumping Lemma, Revisited (Extended Abstract)

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Abstract. We study descriptional and computational properties of the minimal pumping constant induced by the pumping lemma of Stanat and Weiss [D. F. Stanat and S. F. Weiss. A pumping theorem for regular languages. *SIGACT News*, 14(1), 36–37 (Winter 1982)] which provides a necessary and sufficient condition for the regularity of languages.

1 Introduction

Pumping lemmas, e.g., [1,3,8,9,11,12,13,16], are fundamental tools in the study of formal languages and automata theory. Their primary importance lies in their ability to provide necessary conditions for a language to belong to a specific family, such as regular or context-free languages, allowing researchers to prove non-regularity or non-context-freeness through the contrapositive. Beyond proving non-membership, these lemmas offer deep insights into the expressive power and structural limitations of various language-generating or -accepting mechanisms. Furthermore, they have algorithmic utility; for instance, they imply a decision procedure to determine if a regular language is finite or infinite based on the existence of words that exceed the pumping constant.

Recently, research has focused on the descriptional and computational complexity of these lemmas [4,5,6,7], specifically investigating the minimal pumping constant—the smallest value p for which a pumping lemma holds for a given language L. Recent results have established the operational complexity of these constants, determining how they behave under regularity-preserving operations such as the Kleene star, union, intersection, and concatenation. Furthermore, the Pumping-Problem—deciding whether a given value p satisfies a specific lemma for a language represented by an automaton—has been proven to be computationally intractable. While this problem is coNP-complete for deterministic finite automata (DFAs), it becomes PSPACE-complete for nondeterministic finite automata (NFAs), and even undecidable when considering context-free languages under context-free pumping constraints.

A critical distinction exists between pumping lemmas that provide only necessary conditions and those that are both necessary and sufficient. Most classic lemmas, such as those by Kozen [11] or Bar-Hillel [1], are necessary conditions; a language may satisfy the pumping property but still fail to be regular or context-free. In contrast, Jaffe's pumping lemma [9] was the first to provide a necessary and sufficient condition for regularity. The defining characteristic of Jaffe's lemma is that its pumping property respects the Myhill-Nerode equivalence classes of the language, meaning that pumping a word does not change its equivalence class. This allows the lemma to fully characterize the family of regular languages, linking the pumping constant directly to the deterministic state complexity of the language. The pumping lemma proposed by Stanat and Weiss [16] extends these concepts such that pumping respects the classes of the syntactic monoid of the language under consideration. Thus, the pumping constant induced by Stanat and Weiss' pumping lemma is linked to the size of the syntactic monoid of the language.

We investigate Stanat and Weiss' pumping lemma in detail from the perspectives of automata theory and computational complexity. The key findings presented in this paper include—here $\mathtt{mpm}$ refers to the minimal pumping constant induced by the pumping lemma of Stanat and Weiss:

Robustness of the Constant: It is demonstrated that the minimal pumping constant $\mathtt{mpm}(L)$ is robust against variations in the lemma's conditions, such as changing the word length requirement from $|w| \geq p$ to $|w| = p$ or restricting the pumping iterations to a single case ($t = 1$). These findings align with research showing that Jaffe's pumping lemma remains unchanged under similar modifications.

Descriptional Complexity: The study evaluates $\mathtt{mpm}(L)$ by comparing it to other established measures of descriptional complexity. Beside basic facts, specifically, it establishes that $\mathtt{mpm}(L)$ is generally incomparable to both deterministic state complexity ($\mathtt{sc}$) and nondeterministic state complexity ($\mathtt{nsc}$), as there are language families where $\mathtt{mpm}(L)$ is significantly smaller than these measures and others where it is strictly larger. Furthermore, the paper proves that $\mathtt{mpm}(L)$ is always greater than or equal to the constant induced by Jaffe's pumping lemma ($\mathtt{mpe}(L)$).

Structural Bounds: An upper bound for the minimal constant based on the transformation monoid is provided, showing $\mathtt{mpm}(L) \leq \mathtt{llp}(G) + 1$, where $\mathtt{llp}(G)$ refers to the length of the longest path in the directed Cayley graph of the language's transformation monoid. This bound becomes a precise characterization when the transformation monoid is isomorphic to the syntactic monoid. This nicely resembles a similar result on the $\mathtt{mpe}$-relation to the length of the longest path in the DFA obtained in [6], with a subtle difference, because for $\mathtt{mpe}(L)$ the bound on the longest path may not be exact. For a language accepted by an n-state DFA, the $\mathtt{mpm}$-constant is further bounded by $e \cdot n!$, where e is Euler's number.

Computational Intractability: A significant portion of the research focuses on the PUMPING-PROBLEM deciding if a given value p satisfies the lemma for a specific automaton. The paper establishes that this problem is computationally hard: it is PSPACE-complete for NFAs and coNP-complete for DFAs. This complexity is noteworthy because the problem remains intractable even when dealing with deterministic finite automata.

Thus, the obtained results nicely fit to known results for other minimal pumping constants investigated in the literature. In particular, there are strong parallels between the properties of the minimal pumping constant for the Stanat-Weiss pumping lemma and those of other established constants, most notably Jaffe's pumping constant. These similarities manifest across several domains, as mentioned above. Due to space constraints, most proofs are omitted.

2 Preliminaries

A *nondeterministic finite automaton* (NFA) is a quintuple $A = (Q, \Sigma, \cdot, q_0, F)$, where Q is the finite set of *states*, Σ is the finite set of *input symbols*, $q_0 \in Q$ is the *initial state*, $F \subseteq Q$ is the set of *accepting states*, and the *transition function* $\cdot$ maps $Q \times \Sigma$ to 2^Q, where 2^Q refers to the powerset of Q. The *language accepted* by the NFA A is defined as $L(A) = \{\, w \in \Sigma^* \mid (q_0 \cdot w) \cap F \neq \emptyset \,\}$, where the transition function is recursively extended to a mapping $Q \times \Sigma^* \to 2^Q$ in the usual way, namely for $q \in Q$ let $q \cdot \varepsilon = \{q\}$, where ε refers to empty word, and $q \cdot wa = \bigcup_{p \in (q \cdot w)} p \cdot a$, for every $w \in \Sigma^*$ and $a \in \Sigma$. Let $\mathtt{nsc}(L)$ refer to the *nondeterministic state complexity* of L, i.e., the number of states of the minimal nondeterministic finite automaton accepting L.

An NFA A is *deterministic* (DFA) if $q \cdot a$ is a singleton set, for every $q \in Q$ and $a \in \Sigma$. In this case we simply write $q \cdot a = p$ instead of $q \cdot a = \{p\}$. Similarly we define $\mathtt{sc}(L)$ to be the the number of states of the minimal deterministic finite automaton accepting L. We have

$$\mathtt{nsc}(L) \leq \mathtt{sc}(L) \leq 2^{\mathtt{nsc}(L)},$$

for every regular language L. A finite automaton is *minimal* if its number of states is minimal with respect to the accepted language. It is well known that each minimal DFA is isomorphic to the DFA induced by the Myhill-Nerode equivalence relation. The *Myhill-Nerode* equivalence relation $\sim_L$ for a language $L \subset \Sigma^*$ is defined as follows: for $u, v \in \Sigma^*$ let $u \sim_L v$ if and only if $uw \in L \iff vw \in L$, for all $w \in \Sigma^*$. The equivalence class of u is referred to as $[u]_L$ or simply $[u]$ if the language is clear from the context and it is the set of all words that are equivalent to u w.r.t. the relation $\sim_L$, i.e., $[u]_L = \{\, v \mid u \sim_L v \,\}$. Thus $\mathtt{sc}(L)$ is equal to $|\Sigma^*/{\sim_L}|$, which is the number of equivalence classes of $\sim_L$. For nondeterministic finite automata the situation is more involved. Nevertheless, there is a remarkably simple lower bound technique for the nondeterministic state complexity of regular languages, which is due to [2].

Theorem 1. *Let $L \subseteq \Sigma^*$ be a regular language and suppose there exists a set of pairs $S = \{\, (x_i, y_i) \mid 1 \leq i \leq n \,\}$ such that*

1. $x_i y_i \in L$ *for* $1 \leq i \leq n$*,*
2. and $i \neq j$ *implies* $x_i y_j \notin L$ *or* $x_j y_i \notin L$*, for* $1 \leq i, j \leq n$*.*

Then any nondeterministic finite automaton accepting L *has at least* n *states, i.e.,* $n \leq \mathtt{nsc}(L)$*. Here* S *is called an* extended fooling set *of* L*.*

Regular languages satisfy a variety of different pumping lemmas. A necessary and sufficient pumping lemma for regular languages, is due to Jaffe [9]:

Lemma 2 (Jaffe). *A language* L *is regular if and only if there is a constant* p *(depending on* L*) such that the following holds: If* $w \in \Sigma^*$ *and* $|w| = p$*, then there are words* $x \in \Sigma^*$*,* $y \in \Sigma^+$*, and* $z \in \Sigma^*$ *such that* $w = xyz$ *and*[4]

$$wv = xyzv \in L \iff xy^t zv \in L$$

for all $t \geq 0$ *and each* $v \in \Sigma^*$*.*

For a regular language L let $\mathtt{mpe}(L)$ refer to the minimal number p satisfying the conditions of Lemma 2. An alternative version of Lemma 2 appeared in [17, page 86, Lemma 4.3], where the requirement $|w| = p$ is replaced by the condition $|w| \geq p$. In [7] it was shown that this slight change in condition *does not* effect the value of the minimal pumping constant $\mathtt{mpe}(L)$, for a regular language L. Moreover, the following result is also from [7]:

Theorem 3. *Let* L *be a regular language over the alphabet* Σ*. Then*

$$\mathtt{mpe}(L) \leq \mathtt{sc}(L) \leq \sum_{i=0}^{\mathtt{mpe}(L)-1} |\Sigma|^i,$$

while the measures $\mathtt{mpe}$ *and* $\mathtt{nsc}$ *are incomparable in general.*

Further properties of $\mathtt{mpe}$ are: (i) $\mathtt{mpe}(L) = 1$ if and only if $L = \emptyset$ or $L = \Sigma^*$ and (ii) for finite language we have $\ell_{\max}(L) + 1 \leq \mathtt{mpe}(L) \leq \ell_{\max}(L) + 2$, where $\ell_{\max}(L) := \max\{\, |w| \mid w \in L \,\}$, and this range of descriptional complexity can be made more precise. We state the following result without proof, since it is a straightforward rewrite of Jaffe's pumping condition—the details are left to the reader.

Lemma 4. *Let* $L \subseteq \Sigma^*$ *be a finite language. Then* $\mathtt{mpe}(L) = \ell_{\max}(L) + 1$ *if and only if for every word* w *with* $|w| = \ell_{\max}(L) + 1$ *there are words* $x \in \Sigma^*$*,* $y \in \Sigma^+$ *and* $z \in \Sigma^*$ *such that* $w = xyz$ *and*

$$(xz)^{-1} L = \emptyset,$$

where $u^{-1}L := \{\, v \in \Sigma^* \mid uv \in L \,\}$*; otherwise* $\mathtt{mpe}(L) = \ell_{\max}(L) + 2$*.* □

[4] Observe that the words $w = xyz$ and $xy^t z$, for all $t \geq 0$, belong to the same Myhill-Nerode equivalence class of the language L. Thus, one can say that the pumping of the word y in w *respects equivalence classes*.

3 Stanat and Weiss' Pumping Lemma and Variants

As mentioned in the introduction, pumping lemmas generally provide necessary conditions for regularity. However, there are only a few pumping lemmas that are both necessary and sufficient characterizations of regular languages. The most prominent example of such a characterization is Jaffe's pumping lemma [9]. Another characterization, which is less known, is due to Stanat and Weiss [16] and will be discussed below.

Lemma 5 (Stanat & Weiss). *A language $L \subseteq \Sigma^*$ is regular if and only if the following condition holds: there exists a positive integer p such that for every word w of length p or greater, there exist $x, y, z \in \Sigma^*$ with $|y| \geq 1$ such that $w = xyz$ and*

$$uxv \in L \iff uxy^t v \in L,$$

for every $u, v \in \Sigma^*$ and all nonnegative integers $t \geq 0$.

This gives rise to the following definition of a *minimal pumping constant*: for a regular language L, let $\mathtt{mpm}(L)$ denote the minimal number p satisfying the conditions of Lemma 5. As in the case of Jaffe's pumping lemma, where the length condition can be changed from $|w| \geq p$ to $|w| = p$ without changing the minimal pumping constant—see, [7]—such a change by replacing the requirement $|w| \geq p$ by the condition $|w| = p$ results in an alternative version of Lemma 5. It is easy to see that this slight change in condition does not affect the statement. Moreover, it does not affect the minimal pumping constant either. For a regular language L we refer to the minimal number satisfying Lemma 5 with the modified condition $|w| = p$ as $\widetilde{\mathtt{mpm}}(L)$. Obviously, $\widetilde{\mathtt{mpm}}(L) \leq \mathtt{mpm}(L)$. On the other hand, $\mathtt{mpm}(L) \leq \widetilde{\mathtt{mpm}}(L)$, because for every word w that is longer than $\widetilde{\mathtt{mpm}}(L)$ there are words $\widetilde{w} \in \Sigma^*$ and $\widetilde{v} \in \Sigma^*$ with $|\widetilde{w}| = \widetilde{\mathtt{mpm}}(L)$ and $w = \widetilde{w}\widetilde{v}$. But word $\widetilde{w}$ can be pumped, i.e., there are words $\widetilde{x}, \widetilde{y}, \widetilde{z} \in \Sigma^*$ with $|\widetilde{y}| \geq 1$ such that $\widetilde{w} = \widetilde{x}\widetilde{y}\widetilde{z}$ and $u\widetilde{x}v \in L \iff u\widetilde{x}\widetilde{y}^t v \in L$, for every $u, v \in \Sigma^*$ and all nonnegative integers $t \geq 0$. Hence, the word w can be pumped, too. This implies that $\mathtt{mpm}(L) \leq \widetilde{\mathtt{mpm}}(L)$, which proves equality. That is, $\mathtt{mpm}(L) = \widetilde{\mathtt{mpm}}(L)$. Thus, the above given definition of $\mathtt{mpm}(L)$ is robust against this change in the length property.

Moreover, also the pumping property can be simplified to $t = 1$ instead of for every $t \geq 0$, thus leading to a stronger version than the original Stanat and Weiss lemma, without changing the actual minimal pumping constant $\mathtt{mpm}(L)$, for a regular language L.

Lemma 6. *A language $L \subseteq \Sigma^*$ is regular if and only if the following condition holds: there exists a positive integer p such that every word w with $|w| = p$, there exist $x, y, z \in \Sigma^*$ with $|y| \geq 1$ such that $w = xyz$ and*

$$uxv \in L \iff uxyv \in L, \tag{1}$$

for every $u, v \in \Sigma^*$.

Proof. It suffices to show that iterating the equivalence (1) yields

$$uxv \in L \iff uxy^t v \in L$$

for all $u, v \in \Sigma^*$ and all integers $t \geq 0$, which is straight forward, since from $uxv \in L \iff uxyv \in L$ replacing v by yv leads to $uxyv \in L \iff uxy^2v \in L$. Combining the two equivalences gives $uxv \in L \iff uxy^2v \in L$. Repeating this substitution inductively yields the desired original property of Stanat and Weiss. Note that the minimal pumping constant remains the same as in the original pumping lemma, where the condition is required to hold for all $t \geq 0$. □

Note that restricting the number of pumping-iterations to a constant is also possible for Jaffe's pumping lemma like, e.g.,

$$wv = xyzv \in L \iff xzv \in L \tag{2}$$

for each $v \in \Sigma^*$. Here the words w, x, y, and z are required to fulfill the property of Jaffe's pumping lemma, in particular $|w| = p$ is required. Obviously, regularity is equivalent to Jaffe's original pumping lemma which in turn implies the restricted version. Since the restricted version says that there exists p such that for every word w with $|w| = p$, there is a decomposition $w = xyz$ with $y \neq \varepsilon$ such that: $wv \in L \iff xzv \in L$ for each $v \in \Sigma^*$ This means that w and xz are Myhill-Nerode equivalent (same right quotient). Crucially, $|xz| < |w| = p$. So the restricted version of Jaffe's pumping lemma means that every string of length p is Myhill-Nerode equivalent to a strictly shorter string. This means that the Myhill-Nerode classes are all represented by strings of length less than p, implying there are finitely many classes—hence language L is regular. Thus, the restricted version of Jaffe's pumping lemma also implies regularity, and since every regular language satisfies Jaffe's pumping lemma, we have shown all these three conditions, namely, regularity, Jaffe's pumping lemma, and its restricted form are equivalent. Therefore, we get:

Lemma 7. *A language L is regular if and only if there is a constant p (depending on L) such that the following holds: If $w \in \Sigma^*$ and $|w| = p$, then there are words $x \in \Sigma^*$, $y \in \Sigma^+$, and $z \in \Sigma^*$ such that $w = xyz$ and*

$$wv = xyzv \in L \iff xzv \in L$$

for each $v \in \Sigma^*$. □

Since every decomposition witnessing Jaffe's pumping property also witnesses its restricted version, the minimal pumping constant for a regular language L induced by the restricted version is bounded above by $\mathtt{mpe}(L)$. Whether this bound can be strict remains open. Although we have no proof yet, we conjecture that this cannot be the case and thus, both minimal pumping constants coincide.

4 Relation to Other Pumping Constants and State Complexity

First we prove some basic facts for the minimal pumping constant induced by Stanat and Weiss' pumping lemma [16]. From the proof given in [16] we know that

$$\texttt{mpm}(L) \leq \texttt{sc}(L)^{\texttt{sc}(L)}.$$

Later in this section we show that this bound can be significantly improved. We start with the following result that nicely resembles a similar result on the minimal pumping constant induced by Jaffe's pumping lemma [5].

Theorem 8. *Let L be a regular language over the alphabet Σ. Then we have $\texttt{mpm}(L) = 1$ if and only if L is universal or empty, i.e., $L = \Sigma^*$ or $L = \emptyset$.*

As mentioned in the preliminaries, $\ell_{\max}(L)+1 \leq \texttt{mpe}(L) \leq \ell_{\max}(L)+2$ for a finite language L and Lemma 4 provides a more precise characterization of these bounds. In contrast, the case of mpm is slightly simpler, as we see next.

Lemma 9. *Let $L \subseteq \Sigma^*$ be a finite language. Then $\texttt{mpm}(L) = \ell_{\max}(L) + 2$.*

Next we consider the relation to mpe and measures of descriptional complexity of automata such as sc and nsc. We start with the measure mpe, where it is known [5] that $\texttt{mpe}(L) \leq \texttt{sc}(L)$, for every regular language L, and mpe and nsc are incomparable, i.e., (i) there is a family of unary regular languages $(L_i)_{i\geq 2}$ such that the inequality $\texttt{nsc}(L_i) < \texttt{mpe}(L_i)$ holds, and (ii) there is a family of regular languages $(L_i)_{i\geq 4}$, over an alphabet of growing size, such that $3 = \texttt{mpe}(L_i) < i \leq \texttt{nsc}(L_i)$. For the relation between mpe and mpm we find the following situation in general.

Theorem 10. *Let L be any regular language. Then $\texttt{mpe}(L) \leq \texttt{mpm}(L)$.*

Proof. Let p be the minimal pumping constant for the language L w.r.t. Lemma 5. Consider any word w and its decomposition $x, y, z \in \Sigma^*$ with $|y| \geq 1$ such that $w = xyz$ and

$$uxv \in L \iff uxy^tv \in L,$$

for every $u, v \in \Sigma^*$ and all nonnegative integers $t \geq 0$. But then this also holds for $u = \varepsilon$, the empty word, and $v = zv'$, for each $v' \in \Sigma^*$. Thus,

$$uxv = \varepsilon xzv' = xzv' \in L \iff uxy^tv = \varepsilon xy^tzv' = xy^tzv' \in L,$$

all $t \geq 0$ and each $v' \in \Sigma^*$. This in turn is equivalent to

$$wv' = xyzv' \in L \iff xy^tzv' \in L,$$

for all $t \geq 0$ and each $v' \in \Sigma^*$, which is exactly the property of Jaffe's pumping lemma. Therefore, $\texttt{mpe}(L) \leq \texttt{mpm}(L)$ as desired. □

Now what can be said about mpm and the state measures sc and nsc? Both of these measures are incomparable to mpm in general as we shall see next.

Theorem 11. *The following statements hold:*

- *There is a family of regular languages $(L_i)_{i\geq 2}$ such that $\mathtt{mpm}(L_i) < \mathtt{sc}(L_i)$ holds. The same holds true for the nondeterministic state complexity* nsc*.*
- *There is a family of regular languages $(L_i)_{i\geq 2}$ such that $\mathtt{sc}(L_i) < \mathtt{mpm}(L_i)$ holds. The same holds true for the nondeterministic state complexity* nsc*.*

In [6] a nice relation between the measure mpe and the length of the longest path of the automaton, that is, a simple directed path of maximum length from the initial state of the automaton, was shown, namely

$$\mathtt{mpe}(L) \leq \ell_A + 1,$$

where A is a DFA, $L := L(A)$, and ℓ_A refers to the length, i.e., number of transitions, of the longest simple directed path of maximum length in the automaton A starting in the initial state q_0 and ending in any state. While the gap in this inequality can be arbitrarily large, this turns out to be very useful for a restricted class of automata, to be more precise bideterministic finite automata, where the pumping constant can be sandwiched between ℓ_A and $\ell_A + 1$; see [6] for more details.

We can do similar for the measure mpm, by considering the transformation/syntactic monoid of the language accepted by A and the longest path within the associated directed Cayley graph. Let us start with the definition of the transformation/syntactic monoid. The *transformation monoid* $T(A)$ of the DFA $A = (Q, \Sigma, \cdot, q_0, F)$ is the semigroup of all transformations $Q \to Q$ induced by words in Σ^*, i.e.,

$$T(A) = \{\, u^A : Q \to Q \mid u \in \Sigma^* \,\},$$

where u^A is the mapping form $Q \to Q$ which is given by $u^A(q) = q \cdot u$. It should be clear that the mappings $a^A \in \Sigma^A$ for $\Sigma^A = \{\, a^A \mid a \in \Sigma \,\}$ are the *generators* of $T(A)$. These mappings can be described in array form, i.e., a transformation $u^A : Q \to Q$ is written as

$$u^A = \begin{pmatrix} q_0 & q_1 & \dots & q_n \\ u^A(q_0) & u^A(q_1) & \dots & u^A(q_n) \end{pmatrix}$$

if $Q = \{q_0, q_1, \dots, q_n\}$. If the DFA A is minimal, then $T(A)$ is isomorphic to the syntactic monoid $M(L)$ of the language L. The *syntactic monoid* of L is the quotient $M(L) = \Sigma^*/\approx_L$, where the *syntactic congruence* $\approx_L$ is defined over Σ^* by $x \approx_L y$ if and only if $uxv \in L \iff uyv \in L$ for every $u, v \in \Sigma^*$; the *congruence class* of x is referred to as $[x]_{\approx_L}$ or simply $[x]_\approx$ if the language is clear from the context. As above, define $\Sigma_{\approx_L} = \{\, [a]_{\approx_L} \mid a \in \Sigma \,\}$ as *generator set* of $M(L)$. Finally, we define the *directed Cayley graph* $\mathrm{Cay}(M, G)$ of a monoid

$(M, \circ)$ with generator set G as the directed graph whose vertex set is $V := M$ and whose edge set is

$$E := \{\, (m, m \circ g) \mid m \in M \text{ and } g \in G \,\}$$

build by *right multiplication*. If there is no danger of confusion and the set of generators G is clear from the context we simply write $\mathrm{Cay}(M)$ instead of $\mathrm{Cay}(M, G)$. The *length of the longest path* in the Cayley graph $\mathrm{Cay}(M, G)$ is the maximum distance between any two vertices in the graph and it is referred to as $\mathtt{llp}(\mathrm{Cay}(M, G))$. We give a small example.

Example 12. Consider the DFA $A = (Q, \Sigma, \cdot, q_0, F)$ with state set $Q = \{1, 2, 3, 4\}$, alphabet $\Sigma = \{a, b\}$, initial state $q_0 = 1$, and set of final states $F = \{3\}$. The transition function can be read from the automaton drawing shown on the left in Figure 1. The mappings $\varepsilon^A = id$, a^A, b^A, $(aa)^A$, and $(ab)^A$ form the transforma-

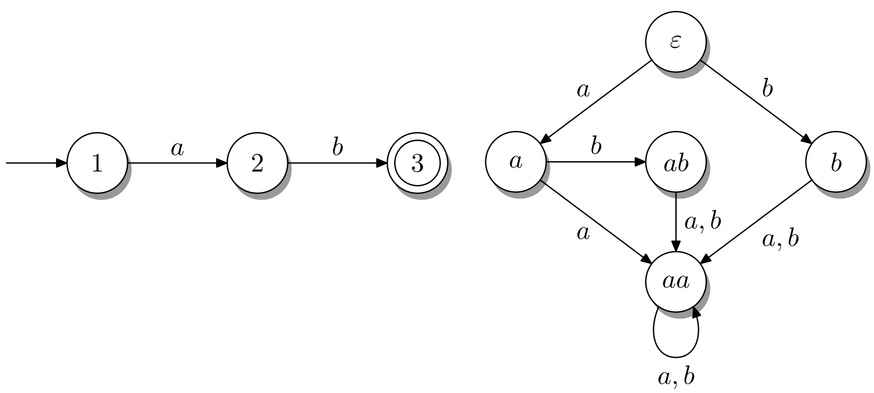

Fig. 1. Left: Minimal DFA A accepting the language $\{ab\}$. The non-accepting sink state is not shown. Right: Cayley graph induced by the transformation monoid of the DFA A, where the superscript A in the vertices and on the edges (labeled by generators) are omitted. In fact $T(A)$ is isomorphic to the syntactic monoid $M(L)$, since A is minimal.

tion monoid $T(A)$, which is isomorphic to the syntactic monoid $M(L)$, since A is minimal. In array form these mapping read as follows:

$$a^A = \begin{pmatrix} 1\,2\,3\,4 \\ 2\,4\,4\,4 \end{pmatrix}, \qquad b^A = \begin{pmatrix} 1\,2\,3\,4 \\ 4\,3\,4\,4 \end{pmatrix},$$

and

$$(aa)^A = \begin{pmatrix} 1\,2\,3\,4 \\ 4\,4\,4\,4 \end{pmatrix}, \qquad (ab)^A = \begin{pmatrix} 1\,2\,3\,4 \\ 3\,4\,4\,4 \end{pmatrix}.$$

The Cayley graph $\mathrm{Cay}(T(A), \Sigma^A)$ or simply $\mathrm{Cay}(T(A))$ induced by the DFA A is depicted on the right in Figure 1. The maximum distance between any of the vertices in $\mathrm{Cay}(T(A))$ is three. There is only one path of maximal length, namely it starts at vertex ε^A, moving in sequence to vertices a^A and $(ab)^A$, and ending in vertex $(aa)^A$. Hence, $\mathtt{llp}(\mathrm{Cay}(T(A))) = 3$. □

Now we are ready for the next theorem, where the proof follows along similar lines as the proof of the pumping lemma by Stanat and Weiss [16]:

Theorem 13. *Let $L \subseteq \Sigma$ be accepted by the DFA $A = (Q, \Sigma, \cdot, q_0, F)$, i.e., $L = L(A)$. Then*

$$\mathtt{mpm}(L) \leq \mathtt{llp}(G) + 1.$$

where G is the Cayley graph $Cay(T(A), \Sigma^A)$ with $\Sigma^A = \{\, a^A : Q \to Q \mid a \in \Sigma \,\}$.

This looks very similar to the result for the measure $\mathtt{mpe}$ mentioned above. Now the question arises, whether the upper bound shown in Theorem 13 is also a lower bound. For arbitrary DFAs, this is generally not the case; but for the minimal finite automaton and the syntactic monoid of the language, the upper bound also serves as a lower bound. Before we can prove this, we recast Stanat and Weiss' pumping condition in terms of the transition function of a DFA accepting L.

Lemma 14. *Let $A = (Q, \Sigma, \cdot, q_0, F)$ be a DFA and let Q_{reach} refer to the set of all reachable states of A, i.e., $Q_{reach} = \{\, q \in Q \mid q = q_0 \cdot w$ for $w \in \Sigma^* \,\}$. Moreover, let $x, y \in \Sigma^*$. Then*

$$uxv \in L(A) \iff uxyv \in L(A) \text{ for every } u, v \in \Sigma^*$$

if and only if

$$(q \cdot x) \cdot y \sim_{L(A)} q \cdot x \text{ for every } q \in Q_{reach}.$$

In case A is the minimal DFA, then the Myhill-Nerode equivalence $\sim_{L(A)}$ is replaced by (ordinary) equality.

Now we are ready for the next theorem:

Theorem 15. *Let $L \subseteq \Sigma^*$ be a regular language with syntactic monoid $M(L)$. Then*

$$\mathtt{mpm}(L) = \mathtt{llp}(G) + 1,$$

where G is the Cayley graph $Cay(M(L), \Sigma_{\approx_L})$ with $\Sigma_{\approx_L} = \{\, [a]_{\approx_L} \mid a \in \Sigma \,\}$.

An easy application of Theorem 15 is given next.

Example 16. We continue Example 12. Recall that $L = \{ab\}$. Then the longest path in the directed Cayley graph induced by the transformation monoid of the minimal DFA accepting L is 3. Thus, by Theorem 15 we have $\mathtt{mpm}(L) = 3+1 = 4$. □

Using Theorem 15 we show how the upper bound on $\mathtt{mpm}(L)$ can be improved. Recall that for a language L accepted by an n-state DFA, it is known from [16] that $\mathtt{mpm}(L) \leq n^n$. To establish the improved bound, we first introduce some additional definitions and notations. Let S_n be the *symmetric group* on n elements, i.e., the group on all permutations on n elements, under composition, and further let T_n be the *full transformation monoid* on n elements, which is the monoid of all mappings on an n-element set under composition. As usual the set of n elements is $\{1, 2, \ldots, n\}$. Obviously $S_n \subseteq T_n$. Notice that S_n is of size $n!$ whereas T_n is of size n^n. For a mapping $t \in T_n$, let the *image* of t be defined as $\mathtt{img}(t) := \{\, t(i) \mid 1 \leq i \leq n \,\}$ and the *rank* of t is the cardinality of its image, i.e., $\mathtt{rank}(t) := |\mathtt{img}(t)|$.

Theorem 17. *For $n \geq 3$ the Cayley graph $Cay(T_n, S)$ has* no *Hamiltonian path for any generating set $S \subseteq T_n$; it follows that the length of the longest path is at most $e \cdot n!$.*

A direct consequence of the previous theorem is the following result on regular languages, which is somehow surprising that although the upper bound on $\mathtt{mpm}$ grows very fast, but n^n grows even faster, i.e., the fraction of $e \cdot n!$ by n^n tends to 0 as n grows to infinity.

Corollary 18. *For $n \geq 3$ it holds $\mathtt{mpm}(L) \leq e \cdot n!$ for any language L accepted by an n-state DFA.* □

Now the question arises whether there is language that is close to the above mentioned bound of $e \cdot n!$. It is known that there is a set of generators of quadratic size containing all transpositions that induces a Hamiltonian path in S_n by applying the Steinhaus-Johnson-Trotter algorithm (see [10, Sec. 7.2.1.2] and [15]) to generate all permutations on n elements in a minimal-change order. In fact, we can improve this further, as shown in the next theorem, by reducing the number of generators to 2—we use the usual cycle notation for permutations.

Theorem 19. *The Cayley graph $Cay(S_n, S)$ with generating set*

$$S = \{\, (1\,2\,\ldots\,n), (n-1\,n) \}$$

of the permutation inducing one cycle on the n elements and the transposition swapping $n-1$ and n while fixing all other elements has a Hamiltonian path for every $n \geq 2$.

As a consequence of the previous theorem we can show the following.

Theorem 20. *For $n \geq 3$ there is a binary language L accepted by an n-state DFA such that $\mathtt{mpe}(L) = n < n! = \mathtt{mpm}(L)$.*

Proof. Let $A = (Q, \Sigma, \cdot, q_0, F)$ with the state set $Q = \{1, 2, \ldots, n\}$, input alphabet $\Sigma = \{a, b\}$, initial state $q_0 = 1$, and set of final states $F = \{n\}$. We define its transition function by setting

$$i \cdot a = \sigma(i) \text{ and } i \cdot b = \tau(i) \text{ for } 1 \leq i \leq n,$$

where $\sigma = (1\,2\,\ldots\,n)$ and $\tau = (n-1\,n)$. Clearly the transformation monoid of A induces the Cayley graph for which we have shown in Theorem 19 that it contains a Hamiltonian path. By Theorem 15 we thus obtain that $\mathtt{mpm}(L(A)) = n!$ since the Cayley graph $\mathrm{Cay}(S_n, S)$ contains exactly $n!$ nodes—one for each permutation in the symmetric group S_n.

Therefore, it remains to show that $\mathtt{mpe}(L(A)) = n$. Observe that the word $w = a^{n-1}$ does not contain a subword that is in $L(A)$ and thus *cannot* be pumped by any of its decompositions w.r.t. Lemma 2 which immediately implies that $\mathtt{mpe}(L(A)) > n-1$. By Theorem 3 we have $\mathtt{mpe}(L(A)) \leq n$ and thus $\mathtt{mpe}(L(A)) = n$. □

5 Computational Complexity of the Pumping-Problem

The Pumping-Problem was introduced in [5] and reads as follows, adapted to Stanat and Weiss' pumping lemma:

Language-Pumping-Problem or for short Pumping-Problem:
Input: a finite automaton A and a natural number p, i.e., an encoding $\langle A, 1^p\rangle$.
Output: Yes, if and only if the statement from Lemma 5 holds for the language $L(A)$ w.r.t. the value p.

In case Jaffe's pumping lemma is considered instead, it was shown in [5] that the problem is coNP-complete for DFAs, while it becomes PSPACE-complete for NFAs. Thus, the Pumping-Problem is a rare example of an automaton problem, which is intractable already for a *single* deterministic finite automaton. Moreover, the minimal pumping constant for Jaffe's pumping lemma cannot be approximated within a factor of $o(n^{1-\delta})$ with $0 \leq \delta \leq 1/2$, for a given n-state NFA, unless the Exponential Time Hypothesis (ETH) fails.

Now let us focus on the Pumping-Problem as defined above with Stanat and Weiss' pumping lemma as basis. We show that from the complexity theoretical perspective the Pumping-Problem has the same complexity as the problem w.r.t. Jaffe's pumping lemma. This is expected for NFAs, but somehow surprising for DFAs, since the number of Myhill-Nerode classes can be significantly smaller then the number of syntactic monoid elements. Nevertheless, we first show that the problem under consideration is contained in PSPACE for NFAs.

Theorem 21. *Given an NFA A and a natural number p, it can be decided in* PSPACE *whether for the language $L(A)$ the statement of Lemma 5 holds for the value p.*

Proof. By our investigation of Stanat and Weiss' pumping lemma and its variants, we have shown that the original pumping lemma is equivalent to its strong version stated in Lemma 6 without altering the pumping constant.

Let $\langle A, 1^p\rangle$ be an input instance of the Pumping-Problem, where the NFA is $A = (Q, \Sigma, \cdot, q_0, F)$. We consider the negation of the pumping condition; since PSPACE is closed under complementation, the result follows. The negated pumping property reads as follows: there is a word $w \in \Sigma^*$ of length $|w| = p$, such that for every $x, y, z \in \Sigma^*$ with $|y| \geq 1$ satisfying $w = xyz$, there exist $u, v \in \Sigma^*$ with ($uxv \notin L(A)$ and $uxyv \in L(A)$) or ($uxv \in L(A)$ and $uxyv \notin L(A)$).

A Turing machine M verifies the negated pumping property as follows: first M guesses a word $w \in \Sigma^p$ and stores it; this uses space $O(n)$, since p is encoded in unary. Next the Turing machine iterates deterministically over all $O(n^2)$ choices of split positions defining a factorization $w = xyz$ with $|y| \geq 1$. For each such pair (x, y) the device M checks that there exist $u, v \in \Sigma^*$ with $uxv \in L(A)$ and $uxyv \notin L(A)$, or *vice versa*. To this end, M first guesses a word u letter-by-letter and simulates A on u, by updating the current reachable subset $S \subseteq Q$ using A's transition function when processing each symbol; thus, $S := \{q_0\} \cdot u$. Here, the current reachable subset is encoded using a bit vector of length n. From S, the Turing machine deterministically computes the subsets

$$S_x := S \cdot x \qquad \text{and} \qquad S_{xy} := S \cdot xy.$$

Now observe that there exists a suffix v with either $uxv \in L(A)$ or $uxyv \in L(A)$ if and only if there exists a word v such that exactly one of the subsets $S_x \cdot v$ and $S_{xy} \cdot v$ intersects F. Whether such a v exists is verified by guessing such a word letter-by-letter and deterministically compute S_{xv} and S_{xyv}. Finally, it remains to verify that exactly one of the computed subsets intersects with F, which again can be done in a deterministic fashion. Since the reachable subsets can be represented in $O(n)$ space and the outer universal loop over the factorizations $w = xyz$ uses only polynomial space, by reusing work space for each factorization, the overall space required is polynomial in n. Thus, it follows that the Pumping-Problem for NFAs is solvable in PSPACE. □

For NFAs it is easy to see that the Pumping-Problem is even PSPACE-complete.

Theorem 22. *Given an NFA A and a natural number p in unary, it is* PSPACE-*complete to decide whether for the language $L(A)$ the statement from Lemma 5 holds for the value p.*

In the remainder of this section, we consider the Pumping-Problem where a DFA together with a number p in unary is given as input. We show coNP-completeness in the next theorem.

Theorem 23. *Given a DFA A and a natural number p, it can be decided in* coNP *whether for the language $L(A)$ the statement of Lemma 5 holds for the value p.*

Thus, it remains to prove coNP-hardness.

Theorem 24. *Given a DFA A and a natural number p in unary, it is* coNP-*hard to decide whether for the language $L(A)$ the statement from Lemma 5 holds for the value p.*

Proof. Given a digraph $G = (V, E)$ on n vertices and a designated vertex $v_0 \in V$, our reduction constructs, in polynomial time, a DFA $A = (Q, \Sigma, \cdot, q_0, \{q_f\})$ such that G has *no* directed Hamiltonian path starting in v_0 if and only if Lemma 5 holds for the value $p = n$.

The construction of the automaton A is as follows: the state set Q consists of the vertex set V, together with a dead state q_d not in V, as well as a single accepting state q_f. The input alphabet is $V \cup \{\, a_v \mid v \in V \,\}$ (the union being disjoint). The transition function on reading $v \in V$ is given by: $u \cdot v = v$ if (u, v) is an arc in E, and $u \cdot v = q_d$ otherwise; the transition function on reading a_v, for vertex $v \in V$, is given by $u \cdot a_v = q_f$ if $u = v$ and $u \cdot a_v = q_d$ otherwise. Moreover, let $q_d \cdot a = q_d$ and $q_f \cdot a = q_f$ for all symbols $a \in \Sigma$, thus effectively obtaining a non-accepting sink state as well as an accepting sink state. All other transitions lead to the dead state q_d. The start state q_0 is the designated vertex v_0. See Figure 2 for a schematic drawing of the construction.

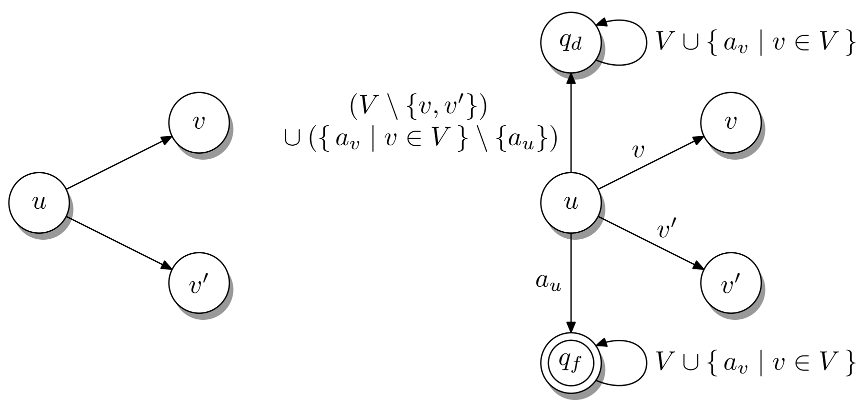

Fig. 2. Left: Schematic drawing of graph $G = (V, E)$ with distinguished vertex v_0 (not shown), depicting vertex u and its edges to v and v'. Right: Schematic drawing of the automaton A with state set $V \cup \{q_d, q_f\}$, input alphabet $V \cup \{\, a_v \mid v \in V \,\}$, initial state v_0 (not shown), and set of final states $\{q_f\}$ constructed from graph G when considering vertex u with its neighboring vertices.

It remains to show that if the graph G has a directed Hamiltonian path starting at v_0, then there is a word w of length n that *cannot* be pumped. Conversely, if no Hamiltonian path exists from v_0, then every word of length n can be pumped, since no such distinguishing structure arises. The proof of these implications are left to the interested reader. □

Thus, we immediately obtain one of the main results of this paper:

Corollary 25. *Given a DFA A and a natural number p in unary, it is* coNP*-complete to decide whether for the language $L(A)$ the statement from Lemma 5 holds for the value p.* □

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